

Deep Deep Learning

Graduate student-led research seminar | Semester: TBD | Course number: TBD

Instructor: Assaf Shocher. Course contact and office hours: TBD.

Course description

Major ideas in deep learning, studied through foundational material, a central paper and recent developments. Topics include NTK and feature learning, Mamba, in-context learning, test-time training, hypernetworks, predictive world models, model merging, interpretability, drifting models and IMLE.

Learning objectives

Explain a mathematical mechanism; identify its assumptions; assess an experiment that tests an explanation; connect recent research to older ideas; formulate a precise unresolved question.

Prerequisites & logistics

Working knowledge of deep learning, linear algebra, probability, calculus and optimization. Formal prerequisites, credits, enrollment, meeting time, room and delivery mode: TBD. There are 13 weekly sessions; dates are not yet assigned. Flow matching is assumed background from the base course.

Seminar format

Students lead presentations and discussion. Each presentation includes one careful derivation, one decisive experiment and one unresolved question. Read selected sections of the background and companion material, with the main paper as the center. Reading sections, assignments and deadlines: TBD.

Grading policy

No exam. Grading components and weights, attendance requirements and any written submissions: TBD.

Research practice & communication

Cite papers, figures, code and other sources; distinguish the authors' claims from your own analysis. Formal academic-integrity and AI-use policies, course forum and accessibility arrangements: TBD.

Weekly schedule

Week numbers only. All student presenters: TBD. TTT is split into adaptation (original Sun et al. and IT3) and neural memory. Drifting models and IMLE have separate sessions.

Week	Topic	Presenter
1	NTK and feature learning	TBD
2	muP, scaling and hyperparameter transfer	TBD
3	State-space models and Mamba	TBD
4	In-context learning as an algorithm	TBD
5	Test-time training: adaptation and IT3	TBD
6	Test-time training: learning as neural memory	TBD
7	Hypernetworks and generated adaptation	TBD
8	JEPA and predictive world models	TBD
9	Model merging and task arithmetic	TBD
10	Mechanistic interpretability: features and superposition	TBD
11	Mechanistic interpretability: circuits and causal tests	TBD
12	Drifting models: training-time distribution dynamics	TBD
13	IMLE and one-step generative models	TBD

Suggested meeting structure

For a 90-minute seminar: 15 minutes foundations, 35 minutes mechanism and derivation, 15 minutes experiment, 25 minutes recent developments and discussion. Actual duration: TBD.

Reading program

Week 1 - NTK and feature learning

When can a neural network be understood as a kernel method, and when do its features change?

Background: Lee et al., [Wide Neural Networks of Any Depth Evolve as Linear Models Under Gradient Descent](#) (2019), selected sections on Taylor linearization and squared-loss dynamics. Review Jacobians, Gram matrices and gradient flow.

Main paper: Jacot, Gabriel & Hongler, [Neural Tangent Kernel: Convergence and Generalization in Neural Networks](#) (2018).

Recent development: Haas et al., [On the Surprising Effectiveness of Large Learning Rates under Standard Width Scaling](#) (2025).

Presentation focus: Derive function-space gradient flow and its solution for squared loss, then compare a finite network with its linearization. Use the recent paper to examine why width, parameterization, learning rate and the choice of loss matter for claims about feature learning.

Week 2 - muP, scaling and hyperparameter transfer

How should initialization and updates scale as networks become wider or deeper?

Background: Yang & Hu, [Tensor Programs IV: Feature Learning in Infinite-Width Neural Networks](#) (2021), selected sections on parameterization and feature learning. The preceding NTK session supplies the contrasting linearized regime.

Main paper: Yang et al., [Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer](#) (NeurIPS 2021; arXiv 2022).

Recent development: Dey et al., [Don't Be Lazy: CompleteP Enables Compute-Efficient Deep Transformers](#) (2025).

Presentation focus: Work through activation and update scaling in a small multilayer network, and inspect a learning-rate transfer experiment. Discuss the additional constraints introduced by depth; width transfer alone does not establish depth transfer or effective nonlinear learning.

Week 3 - State-space models and Mamba

How do recurrence, selectivity and structured matrices produce an efficient sequence model?

Background: Gu, Goel & Ré, [Efficiently Modeling Long Sequences with Structured State Spaces](#) (S4, 2021/22), selected sections on discretization and recurrence. Gu & Dao, [Mamba: Linear-Time Sequence Modeling with Selective State Spaces](#) (2023), selected sections on input-dependent dynamics.

Main paper: Dao & Gu, [Transformers Are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality](#) (Mamba-2, 2024).

Recent development: Lahoti et al., [Mamba-3: Improved Sequence Modeling Using State Space Principles](#) (2026).

Presentation focus: Unroll an SSM into a structured matrix and explain the restricted attention-SSM duality used by Mamba-2. Examine a state-tracking experiment from Mamba-3 and distinguish algorithmic complexity from measured hardware efficiency.

Week 4 - In-context learning as an algorithm

How can a forward pass perform learning without changing the network's parameters?

Background: von Oswald et al., [Transformers Learn In-Context by Gradient Descent](#) (2023), selected sections on the linear-attention construction. Review least squares, ridge regression and Bayesian prediction for linear models.

Main paper: Akyürek et al., [What Learning Algorithm Is In-Context Learning? Investigations with Linear Models](#) (2023).

Recent development: Wurgajt et al., [In-Context Learning Strategies Emerge Rationally](#) (2025).

Presentation focus: Work through one regression-learning construction and compare gradient-descent, ridge and Bayesian predictors on a controlled task distribution. Separate existence constructions and observed behavior in trained synthetic-task models from claims about general language models.

Week 5 - Test-time training: adaptation and IT3

What unlabeled objective makes adaptation improve the task we actually care about?

Background: Sun, Wang, Liu, Miller, Efros & Hardt, [Test-Time Training with Self-Supervision for Generalization under Distribution Shifts](#) (ICML 2020). Shared encoders, self-supervised auxiliary tasks and the gradient-alignment condition under the paper's smooth/convex assumptions.

Main paper: Durasov, Shocher, Oner, Chechik, Efros & Fua, [IT3: Idempotent Test-Time Training](#) (ICML 2025). Learn prediction feedback and use consistency between successive predictions as the test-time objective.

Recent development: IT3 is the recent anchor. Supplementary resources: [arXiv paper](#) and [official implementation](#). Compare its idempotence-based consistency objective with the original TTT auxiliary-task design.

Presentation focus: Derive the original gradient-alignment argument and IT3 consistency loss, then inspect an adaptation experiment and a failure case. IT3 requires its own training setup; discuss when consistency can be uninformative and how to stop adaptation.

Week 6 - Test-time training: learning as neural memory

How can parameter updates store context, and how should training prepare a model to learn at inference?

Background: Ba et al., [Using Fast Weights to Attend to the Recent Past](#) (2016). Fast versus slow weights and associative memory; recap the original Sun et al. TTT formulation.

Main paper: Sun et al., [Learning to \(Learn at Test Time\): RNNs with Expressive Hidden States](#) (2024). A model is the recurrent hidden state, and self-supervised gradient updates are the state transition.

Recent development: Tandon et al., [End-to-End Test-Time Training for Long Context](#) (2025). Next-token updates compress context into weights, with a meta-learned initialization.

Presentation focus: Work through a TTT-Linear update and its recurrent computation. Compare layer-local reconstruction with end-to-end next-token adaptation; distinguish sequence memory from adapting to a shifted test distribution.

Week 7 - Hypernetworks and generated adaptation

Can a network generate the parameters another network needs for a new task or context?

Background: Ha, Dai & Le, [HyperNetworks](#) (2016/17). Hu et al., [LoRA: Low-Rank Adaptation of Large Language Models](#) (2021/22), selected sections on low-rank parameter updates.

Main paper: Charakorn et al., [Text-to-LoRA: Instant Transformer Adaption](#) (2025).

Recent development: Charakorn et al., [Doc-to-LoRA: Learning to Instantly Internalize Contexts](#) (2026).

Presentation focus: Explain the conditioning signal, generated low-rank update and training objective, then examine a held-out-task or document experiment. Compare the cost and information limits of generated adaptation with fine-tuning and keeping the original context available.

Week 8 - JEPA and predictive world models

When does prediction in representation space support understanding, dynamics and planning?

Background: Assran et al., [Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture \(I-JEPA\)](#) (2023), selected sections on masking, target representations and collapse prevention.

Main paper: Assran et al., [V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning](#) (2025).

Recent development: Maes et al., [LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels](#) (2026).

Presentation focus: Trace the representation-prediction objective and action-conditioned planning loop, then analyze a planning result and its data requirements. Contrast target-network stabilization with regularizing the latent distribution; representation quality and accurate action-conditioned dynamics are separate questions.

Week 9 - Model merging and task arithmetic

Why can adding parameter updates combine capabilities, and when does task interference prevent it?

Background: Ilharco et al., [Editing Models with Task Arithmetic](#) (2023). Review parameter differences relative to a shared pretrained checkpoint and the NTK session's function-space linearization.

Main paper: Ortiz-Jimenez, Favero & Frossard, [Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models](#) (2023).

Recent development: Hitit, Girrbaach & Akata, [A Systematic Study of In-the-Wild Model Merging for Large Language Models](#) (2025 preprint; TMLR 2026).

Presentation focus: Derive how adding task vectors affects a linearized model and examine the proposed weight-disentanglement mechanism. Compare controlled task combinations with heterogeneous checkpoints, including conflicting objectives and gains relative to the best individual model.

Week 10 - Mechanistic interpretability: features and superposition

What is a feature, and can sparse autoencoders recover the units a network computes with?

Background: Elhage et al., [Toy Models of Superposition](#) (2022), selected sections on sparse features, interference and the two-feature toy model.

Main paper: Templeton et al., [Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet](#) (2024).

Recent development: Leask et al., [Sparse Autoencoders Do Not Find Canonical Units of Analysis](#) (2025).

Presentation focus: Explain the sparse-autoencoder objective and reproduce or analyze a small superposition experiment. Examine feature interventions alongside the stitching and meta-SAE evidence, asking whether interpretability, completeness and atomicity are being measured separately.

Week 11 - Mechanistic interpretability: circuits and causal tests

How can we trace a model's computation and test whether the proposed explanation is causal?

Background: Elhage et al., [A Mathematical Framework for Transformer Circuits](#) (2021), selected sections on the residual stream and attention's QK/OV decomposition. The preceding features session supplies sparse representations.

Main paper: Ameisen et al., [Circuit Tracing: Revealing Computational Graphs in Language Models](#) (2025).

Recent development: Kamath et al., [Tracing Attention Computation Through Feature Interactions](#) (2025).

Presentation focus: Explain how a replacement model produces an attribution graph and work through one circuit case study with an intervention. Identify approximation errors and omitted computations, then examine what the attention extension adds to the explanation.

Week 12 - Drifting models: training-time distribution dynamics

How can attraction and repulsion during training produce a one-pass generator, and when does zero drift identify the correct distribution?

Background: Yujia Li, Swersky & Zemel, [Generative Moment Matching Networks](#) (ICML 2015). Kernel mean embeddings, maximum mean discrepancy and distribution matching from samples.

Main paper: Deng, He Li, Tianhong Li, Du & He, [Generative Modeling via Drifting](#) (2026). Estimate an attraction-repulsion field from real and generated samples and fit frozen drifted targets.

Recent development: Esteban-Casadevall et al., [Kernel-Gradient Drifting Models](#) (2026), including Max Welling. A mathematical companion for kernel-gradient fields, smoothing and distribution identifiability. [Original drifting project and demonstrations](#).

Presentation focus: Derive the drift and regression update, including the role of stopping gradients through the target. Examine a 2D example, equilibrium assumptions and one ablation; arbitrary zero drift need not certify distribution equality.

Week 13 - IMLE and one-step generative models

Can data-to-generated nearest-neighbor matching achieve coverage and high fidelity without an iterative sampler?

Background: Ke Li & Jitendra Malik, [Implicit Maximum Likelihood Estimation](#) (2018). Implicit distributions, nearest-neighbor assignment, mode coverage and the likelihood connection under stated assumptions.

Main paper: Chirag Vashist & Ke Li, [ROMS-IMLE: A Minimalist Approach to Competitive Single-Step Generative Modelling](#) (2026). Revisit IMLE with intermediate-resolution supervision and robust losses.

Recent development: ROMS-IMLE is the recent anchor. The authors' [method page](#) provides additional explanation and experiments.

Presentation focus: Explain the direction of nearest-neighbor matching, latent assignment and alternating regression, then examine sampling cost and the ROMS-IMLE ablations. Keep the maximum-likelihood connection tied to its assumptions.

Optional substitutions

These complete reading bundles can replace a scheduled session after coordinating with the instructor. Flow matching itself was covered in the base course.

Diffusion language models

How can denoising generate discrete text, and what changes when generation need not proceed left to right?

Background: Austin et al., [Structured Denoising Diffusion Models in Discrete State-Spaces](#) (2021), selected sections on transition matrices and absorbing-mask corruption.

Main paper: Sahoo et al., [Simple and Effective Masked Diffusion Language Models](#) (2024).

Recent development: Nie et al., [Large Language Diffusion Models](#) (LLaDA, 2025).

Presentation focus: Derive the masked-denoising objective and trace one sampling schedule, then compare refinement, parallelism and likelihood evaluation with autoregressive generation. The topic requires discrete corruption and language-specific evaluation beyond a continuous diffusion refresher.

Geometry of optimization: Muon

How does the norm used to measure an update change the direction of steepest descent?

Background: Bernstein & Newhouse, [Old Optimizer, New Norm: An Anthology](#) (2024), selected sections on steepest descent, dual norms and matrix operator norms.

Main paper: Liu et al., [Muon Is Scalable for LLM Training](#) (2025).

Recent development: Kovalev, [Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization](#) (2025).

Presentation focus: Derive the spectral-norm-constrained linearized update using the SVD and explain its polar factor and Newton-Schulz approximation. Examine a scaling experiment and distinguish the geometric argument from implementation choices, weight decay and empirical comparisons with AdamW.

RL for reasoning with verifiable rewards

Does reward-based training discover successful reasoning behavior or change which existing behavior is sampled?

Background: [Spinning Up: Part 3-Introduction to Policy Optimization](#), selected sections on the log-derivative policy gradient and baselines. Reserve 25-30 minutes to introduce policies, sampled completions, rewards and group-relative advantages; prior RL coursework is not assumed.

Main paper: DeepSeek-AI et al., [DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning](#) (2025), focusing on R1-Zero and the learning objective.

Recent development: Yue et al., [Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?](#) (2025). Counterpoint: Liu et al., [ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models](#) (2025).

Presentation focus: Derive the policy-gradient identity before introducing group-relative advantages, clipping and KL regularization. Compare pass@k evidence under the papers' different training and evaluation settings; finite sampling cannot establish that a base model assigns zero probability to a solution.

Course website: [Deep Deep Learning](#)